



JUNE 2023 EXAMINATION

MODULE DESCRIPTION	:	DEVELOPMENT SOFTWARE IV
MODULE CODE	:	ONT401
FACULTY	:	ENGINEERING, THE BUILT ENVIRONMENT AND IT
QUALIFICATION	:	Adv. Diploma INFORMATION TECHNOLOGY
EXAMINATION DATE	:	 JUNE 2023
SESSION STARTS AT	:	09:00 <u>OR</u> 14:00
DURATION (IN MINUTES)	:	180
TOTAL MARKS	:	100
PAGES	:	4
ADDENDA	:	NONE
EXAMINER	:	Mr VS Mdunyelwa
MODERATOR	:	MR R WANVIG
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. CODE MUST BE WRITTEN IN PEN. 3. YOU MAY KEEP THE QUESTION PAPER.
REQUIREMENTS	:	NO CALCULATORS ARE REQUIRED, BUT THEY MAY BE USED IF DESIRED

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QUESTION 1 [20]

Roboco are in the business of building robots to aid in everyday tasks for both the residential and commercial sectors. They have heard that the concepts of the strategy pattern and delegates might help them create flexible designs for their robots. They have contacted you to devise a proof of concept using C# and a console application. They provided the following specifications for their robots:

- Every **Robot** needs to have a string-based model number.
- Every **Robot** requires a type of hand. They want to group the different hands using the term **HandType**. The **HandType** should be used to demonstrate the strategy pattern using composition. Every hand should be able to **Grasp** an object. When an object is **Grasped**, an appropriate message should be displayed to the console window. They currently build robots with 3 types of hands:
 - **Human**: Grasps things like a normal human hand.
 - **SuctionCup**: Works by sucking onto the object – this kind of hand is good for moving things like panes of glass.
 - **Pincer**: This type of hand has 2 metal fingers. This is the simplest / cheapest kind of hand the company produces.
- Every **Robot** requires a type of vision (how they see). They want to group the different vision types using the term **VisionType**. The **VisionType** should be used to demonstrate the strategy pattern using delegates. Every vision type should be able to **See** an object. When an object is **Seen**, an appropriate message should be displayed to the console window. They currently build robots with 3 types of vision:
 - **VisibleLight**: Sees things like a human would.
 - **UltraViolet**: Sees in the ultraviolet spectrum.
 - **Predator**: Sees in the infrared spectrum; kind of like the monster from the movie(s) Predator.

In the **Main** method of your program, your prototype should:

- Create at least 3 different Robots with different Hands and types of Vision.
- Demonstrate that each Robot is able to Grasp and See.
- Change the Hand and Vision type of one of the Robots to demonstrate the interchangeability of the behaviours.

To answer this question, you are required to write down a code segment for each of your answers to the following 4 questions. You start each question with full marks, but marks are deducted for omissions, deviations and mistakes.

1.1 Write down all the code related to HandType. (5)

1.2 Write down all the code related to VisionType. (5)

1.3 Write down the code for class Robot. (5)

1.4 Write down the code which would appear in the Program class' Main method. (5)

[20]

Question 2 [38]

You have been employed to work in a company which develops animal characters for a television set. The animals need to have the capability of making sounds. The animals which are available as characters are a Cat and a Dog. You are required to simulate the sounds of the animals by returning the relevant sounds which the animals make. The animals should be provided with their sounds and allowed to make their sounds when the program is played.

You are required to use your knowledge of the Abstract Factory design pattern to answer the following questions.

- 2.1 Define the abstract factory design pattern and align it to the scenario. (5)
- 2.2 Draw a class diagram suitable for the scenario with the necessary methods and the relationships between the classes. (5)
- 2.3 You are required to write the code for:
 - 2.3.1 The IAnimal interface. (4)
 - 2.3.2 The Dog and Cat classes. (6)
 - 2.3.3 The IAnimalFactory. (6)
 - 2.3.4 The Dog and Cat factory classes. (4)
 - 2.3.5 The Client class. (4)
 - 2.3.6 The Main program which allows creating a cat and dog characters. (4)

[38]**QUESTION 3 [30]**

For the following questions, you are required to:

- Provide a definition of each design pattern using different examples which align to the design patterns.
- Draw a template class diagram for each design pattern.
- Write down the template code for each design pattern (default, generic class pattern and method names) for the pattern specified. This must include the code for all the interfaces and classes as required for the default template. Also include a section of code from the program's Main method to demonstrate how to use the implemented pattern.

- 3.1 Decorator Pattern (15)
- 3.2 Command Pattern (15)

[30]**Question 4 [12]**

The Singleton Pattern ensures a class has only one instance, and provides a global point of access to it. Many implementations for such a singleton exists, one of which is called the double locking approach. Write the code for a singleton class using this approach and comment each line of code to clearly explain every single keyword/feature in the code.

(Note: Marks will only be allocated for extensively commented code that clearly explains both the meaning of the keywords used and the reason for using them in the specific context. Perfect code with unclear comments is worth no marks! You have to demonstrate that you understand the code.)

[12]